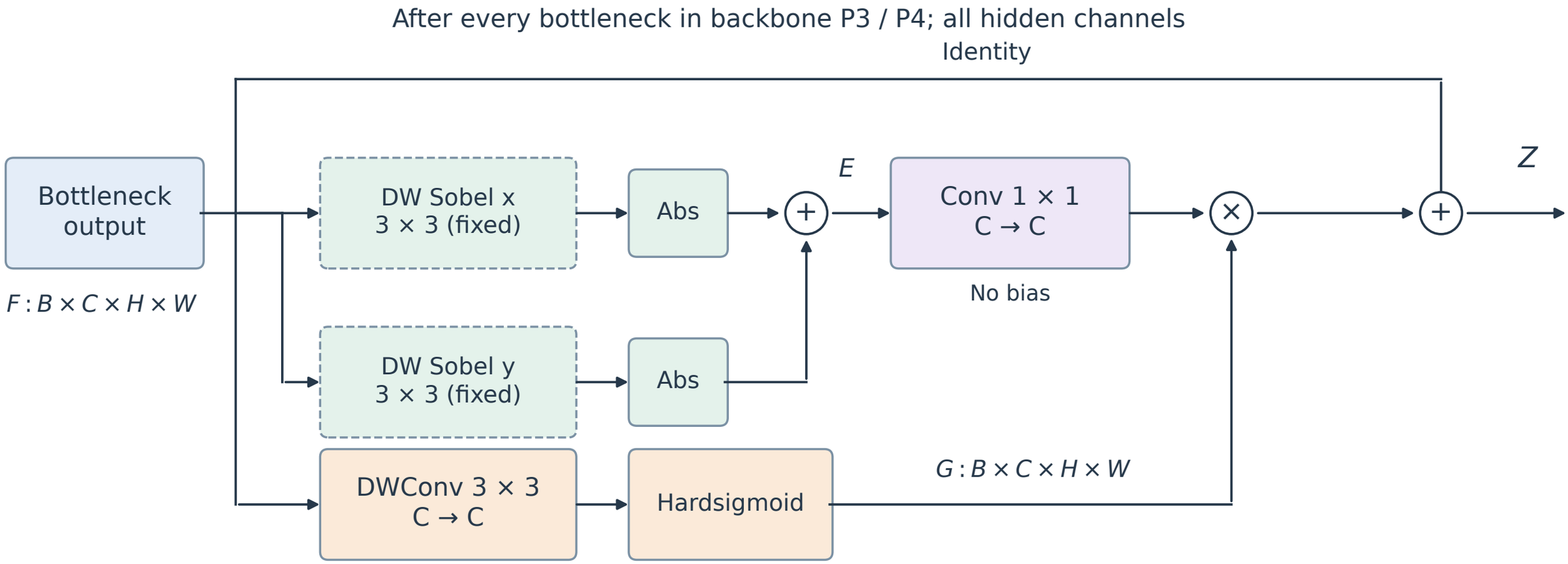
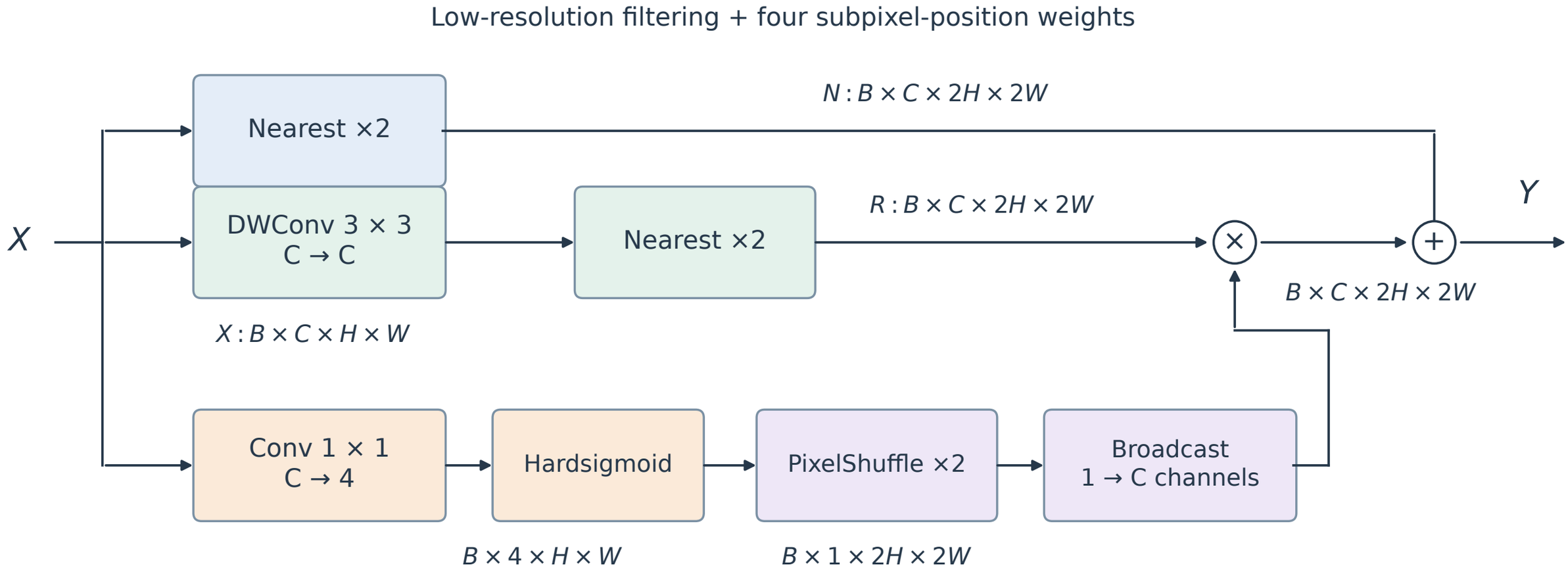
(a) SGRE | Selective Gradient Residual Enhancement



$$Z = F + \mathrm{HS}(\mathrm{DWConv}_{3\times3}(F)) \odot \mathrm{Conv}_{1\times1}(|K_x *_{\mathrm{dw}} F| + |K_y *_{\mathrm{dw}} F|)$$

(b) SARU | Subpixel-Adaptive Residual Upsampling



$$Y = \mathrm{Nearest}_2(X) + \mathrm{PS}_2(\mathrm{HS}(\mathrm{Conv}_{1\times1}(X))) \odot \mathrm{Nearest}_2(\mathrm{DWConv}_{3\times3}(X))$$